3 A spring obeying Hooke's law has an unstretched length of 45 mm and a spring constant of 380 N m^{-1}.

What is the increase in the elastic potential energy of the spring when its length increases from 50 mm to 60 mm?

A 0.019 J **B** 0.038 J **C** 0.076 J **D** 0.21 J

Ans: **B**

Extension goes from 5 mm to 15 mm.

$$\text{increase in EPE} = \frac{1}{2}k\left(x_f^2 - x_i^2\right) = \frac{1}{2}(380)\left(0.015^2 - 0.005^2\right)$$
$$= 0.038 \text{ J}$$

A: use $(1/2)k(\Delta x)^2$; C: forgot × ½ ; D: took length instead of extension